MATH 128a: Homework #9

Instructor: F. Weber

Due by: **Wednesday, November 5, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 3.2

Solve problems 6, 7 of Section 3.2 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 2: Divided Differences Matlab

(a) Implement Newton's divided differences formula (Algorithm 3.2 in the textbook) in Matlab.

(b) Implement a matlab function that evaluates Newton's polynomial at a given point x given interpolation points $x_0, \ldots, x_n$ and the divided differences from (a).

Problem 3: Divided differences

Prove that $f[x_1, \ldots, x_k, x_0] = f[x_0, x_1, \ldots, x_k]$ for any $k \geq 1$.

Problem 4: Problems from the textbook, chapter 3.3

Solve problems 2, 4, 7, 14(16), 19(10), 22(20) of Section 3.3 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 5: Hermite Interpolation

Implement a function in matlab `[coeffs] = hermite(xi,fi,dfi)` that returns the coefficients of the Hermite polynomial using divided differences interpolating at values `(xi,fi,dfi)` (where `xi` are the values x_i `fi` are the values $f(x_i)$ and `dfi` are the values of the derivatives $f'(x_i)$).